Anthony J Barrows

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Education

PhD in Complex Systems and Data Science — 2022-2026 (Expected)
MS in Complex Systems and Data Science — 2019-2021
The University of Vermont, Burlington, Vermont

BA in Psychology — 2013–2016
University of Hartford, Hartford, Connecticut
Honors Thesis: *Characteristics Associated with MMPI-2 L-scale Elevations in a Sample of Police Officer Candidates*

Experience

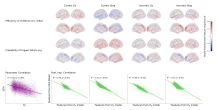
Teaching Fellow, ABCD-ReproNim Course — 2025-2026
Writing Consultant, Graduate Writing Center — 2023–Present
Program Representative, Graduate Student Senate — 2023–2024

Research Data Analyst, Vermont Center on Behavior and Health — 2021–2022
Lead Research Assistant, Vermont Center on Behavior and Health — 2019–2021
Research Assistant, Vermont Center on Behavior and Health — 2016–2019

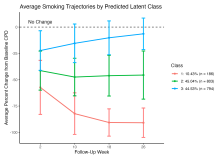
Awards

NEuroimaging Reserach VErmont (NERVE) Lab predoctoral fellow (T32 DA043593)
Harry L. Leonhard Award for Greatest Contribution to the Psychology Department by a Graduating Psychology Major

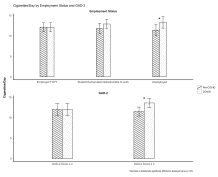
Publications



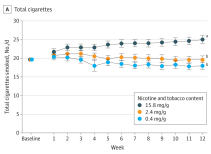
6. **Barrows, A.**, Weigard, A., McCabe, M., Potter, A., Garavan, H., and Allgaier, N. (2025). Cognitive process models reveal meaningful brain-behavior associations in the abcd study stop-signal task. *Under review at Journal of Cogitive Neuroscience*



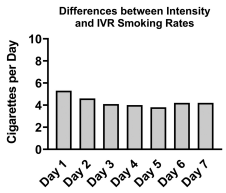
5. **Barrows, A.**, Klemperer, E., Garavan, H., Allgaier, N., Lindson, N., and Taylor, G. (In Press). Smoking reduction trajectories, and their association with smoking cessation: A secondary analysis of longitudinal RCT data. *BMJ Public Health*



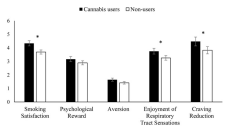
4. Wiley, R. C., Oliver, A. C., Snow, M. B., Bunn, J. Y., **Barrows, A. J.**, Tidey, J. W., Lee, D. C., Sigmon, S. C., Gaalema, D. E., Heil, S. H., Markesich, C., Villanti, A. C., and Higgins, S. T. (2021). The Impact of the Covid-19 Pandemic on Smoking Among Vulnerable Populations. *Nicotine & Tobacco Research*



3. Higgins, S. T., Tidey, J. W., Sigmon, S. C., Heil, S. H., Gaalema, D. E., Lee, D., Hughes, J. R., Villanti, A. C., Bunn, J. Y., Davis, D. R., Bergeria, C. L., Streck, J. M., Parker, M. A., Miller, M. E., DeSarno, M., Priest, J. S., Cioe, P., MacLeod, D., **Barrows, A. J.**, Markesich, C., and Harfmann, R. F. (2020). Changes in Cigarette Consumption With Reduced Nicotine Content Cigarettes Among Smokers With Psychiatric Conditions or Socioeconomic Disadvantage: 3 Randomized Clinical Trials. *JAMA Network Open*, 3(10):e2019311–e2019311



2. Nighbor, T. D., **Barrows, A. J.**, Bunn, J. Y., DeSarno, M. J., Oliver, A. C., Coleman, S. R. M., Davis, D. R., Streck, J. M., Reed, E. N., Reed, D. D., and Higgins, S. T. (2020). Comparing Participant Estimated Demand Intensity on the Cigarette Purchase Task to Consumption when Usual-Brand Cigarettes were Provided Free. *Preventive Medicine*



1. Parker, M. A., Streck, J. M., Bergeria, C. L., Bunn, J. Y., Gaalema, D. E., Davis, D. R., **Barrows, A. J.**, Sigmon, S. C., Tidey, J. W., Heil, S. H., and Higgins, S. T. (2018). Reduced Nicotine Content Cigarettes and Cannabis Use in Vulnerable Populations. *Tobacco Regulatory Science*, 4(5):94–91

Talks

2. **Barrows, A. J.** (2023). Improving brain-behavior associations by strengthening models of behavior (Talk). *Adolescent Brain Cognitive Delopment (ABCD) Study 2023 Annual Meeting*

1. Taylor, S., Garavan, H., Weigard, A., Ging-Jehli, N., Hitchcock, P., and **Barrows, A. J.** (2023). What can computational modeling reveal about the generality or specificity of neurocognitive risk factors for psychopathology? (Panel Presentation). *American College of Neuropsychopharmacology 2023 Annual Meeting*

Posters

5. **Barrows, A. J.**, Pearson, E., Chaarani, B., Garavan, H., and Allgaier, N. (2023). Longitudinal task activation differences for the MID task in the ABCD Study (Poster Presentation). *Organization for Human Brain Mapping 2023 Annual Meeting*

4. Wiley, R. C., Oliver, A. C., Bunn, J. Y., **Barrows, A. J.**, Tidey, J. W., Lee, D., Sigmon, S., Gaalema, D. E., Heil, S. H., Markesich, C., Villanti, A. C., and Higgins, S. T. (2020a). Urban and rural differences in smoking behavior among vulnerable populations during the COVID-19 pandemic (Poster Presentation). *2020 Vermont Center on Behavior and Health Conference*

3. Wiley, R. C., Oliver, A. C., Snow, M. B., Bunn, J. Y., **Barrows, A. J.**, Tidey, J. W., Lee, D., Sigmon, S., Gaalema, D. E., Heil, S. H., Markesich, C., Villanti, A. C., and Higgins, S. T. (2020b). Smoking behavior among vulnerable populations in response to the COVID-19 pandemic (Abstract). *2020 Tobacco Centers of Regulatory Science Grantee Meeting*

2. Streck, J. M., Sigmon, S. C., Parker, M. A., Stitzer, M. L., Heil, S. H., Tidey, J. W., Gaalema, D. E., Bergeria, C. L., Davis, D. R., Priest, J., **Barrows, A. J.**, Rothman, M., Hughes, J. R., and Higgins, S. T. (2017). Subjective effects of smoking among opioid-maintained individuals: Results from a trial examining reduced nicotine content cigarettes (Abstract). *The College on Problems of Drug Dependence 2018 Annual Meeting*

1. **Barrows, A. J.** (2016). Personality characteristics associated with MMPI-2 L-Scale elevations (Poster Presentation). *2016 Eastern Psychological Association Annual Meeting*

Computer Skills

Programming Languages: R, Python, LaTeX, Markdown
Frameworks: R Shiny, R Markdown, Microsoft Access, Linux CLI, REDCap API
Developer Tools: R Studio, Jupyter, Visual Studio Code
Version Control: Git, GitHub
Libraries: Nilearn, tidyverse, ggplot2, lcmm, Plotly, pandas, NumPy, Matplotlib, scikit-learn
Software: Microsoft Excel, REDCap (Project Design)